

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE

SUBJECT : Cyber & Information Security

Aim: Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

Input

```
# Diffie-Hellman Key Exchange

# Public values
p = int(input("Enter a prime number (p): "))
g = int(input("Enter a primitive root (g): "))

# Private keys
aarav_private = int(input("Enter Aarav's private key: "))
vihaan_private = int(input("Enter Vihaan's private key: "))

# Calculate public keys
aarav_public = (g ** aarav_private) % p
vihaan_public = (g ** vihaan_private) % p

print("\nAarav's Public Key:", aarav_public)
print("Vihaan's Public Key:", vihaan_public)

# Calculate shared secret keys
aarav_shared = (vihaan_public ** aarav_private) % p
vihaan_shared = (aarav_public ** vihaan_private) % p

print("\nAarav's Shared Secret Key:", aarav_shared)
print("Vihaan's Shared Secret Key:", vihaan_shared)

# Check whether both keys are same
if aarav_shared == vihaan_shared:
    print("\nKey Exchange Successful!")
    print("Both Aarav and Vihaan have the same secret key.")
else:
    print("\nKey Exchange Failed!")
```

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Logic of the Code

The logic of the program is based on modular exponentiation.

For Aarav:

$\text{Aarav Public Key} = (g ^ \text{Aarav Private Key}) \bmod p$

For Vihaan:

$\text{Vihaan Public Key} = (g ^ \text{Vihaan Private Key}) \bmod p$

After exchanging the public keys:

$\text{Aarav Shared Key} = (\text{Vihaan Public Key} ^ \text{Aarav Private Key}) \bmod p$

$\text{Vihaan Shared Key} = (\text{Aarav Public Key} ^ \text{Vihaan Private Key}) \bmod p$

Both calculations produce the same value. The program compares these two values to check whether the key exchange was successful.

Functions and Syntax Used

`input()` is used to accept values from the user.

`int()` converts the input value into an integer.

`print()` is used to display the public keys, shared keys and result.

`**` is the exponentiation operator used to calculate the power of a number.

`%` is the modulus operator used to calculate the remainder.

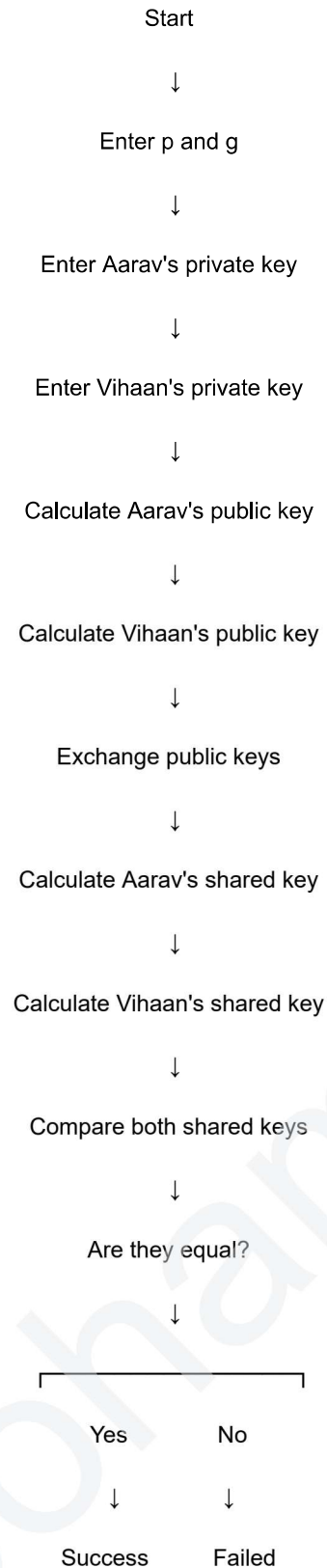
`=` is the assignment operator used to store values in variables.

`==` is the comparison operator used to compare the shared keys.

`if-else` is used to check whether both shared secret keys are equal.

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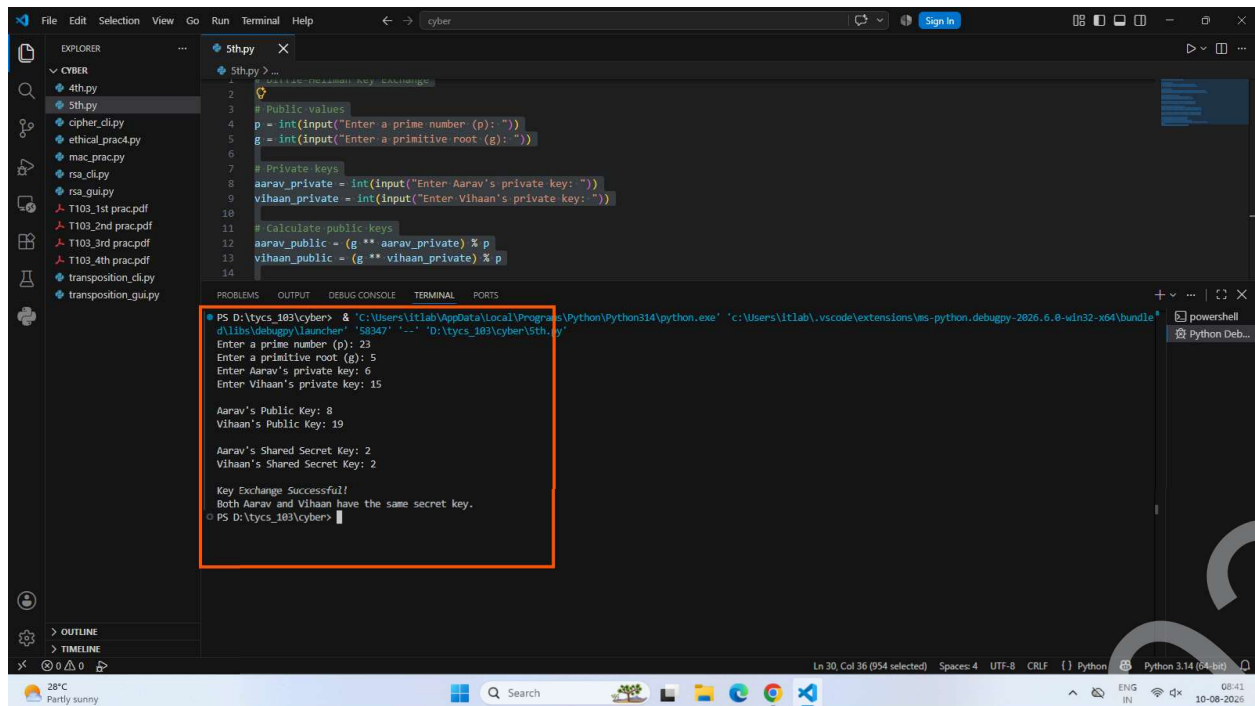
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Output



The screenshot shows a VS Code editor with a file explorer on the left containing files like 4th.py, 5th.py, cipher_cli.py, ethical_prac4.py, mac_prac.py, rsa_cli.py, rsa_gui.py, T103_1st_prac.pdf, T103_2nd_prac.pdf, T103_3rd_prac.pdf, T103_4th_prac.pdf, transposition_cli.py, and transposition_gui.py. The main editor displays a Python script named 5th.py. The terminal at the bottom shows the execution of the script, which prompts for a prime number (p), a primitive root (g), and private keys for Aarav and Vihaan. It then calculates their public keys and a shared secret key, confirming a successful key exchange.

```
5th.py
1 # Diffie-Hellman Key Exchange
2
3 # Public values
4 p = int(input("Enter a prime number (p): "))
5 g = int(input("Enter a primitive root (g): "))
6
7 # Private keys
8 aarav_private = int(input("Enter Aarav's private key: "))
9 vihaan_private = int(input("Enter Vihaan's private key: "))
10
11 # Calculate public keys
12 aarav_public = (g ** aarav_private) % p
13 vihaan_public = (g ** vihaan_private) % p
14
15 # Calculate shared secret key
16 aarav_shared_secret = (vihaan_public ** aarav_private) % p
17 vihaan_shared_secret = (aarav_public ** vihaan_private) % p
18
19 # Verify the shared secret key
20 if aarav_shared_secret == vihaan_shared_secret:
21     print("Key Exchange Successful!")
22 else:
23     print("Key Exchange Failed!")
```

PS D:\tycs_103\cyber> .\5th.py
Enter a prime number (p): 23
Enter a primitive root (g): 5
Enter Aarav's private key: 6
Enter Vihaan's private key: 15

Aarav's Public Key: 8
Vihaan's Public Key: 19

Aarav's Shared Secret Key: 2
Vihaan's Shared Secret Key: 2

Key Exchange Successful!
Both Aarav and Vihaan have the same secret key.
PS D:\tycs_103\cyber>

Gui Input

```
import tkinter as tk
from tkinter import messagebox

# ----- Diffie-Hellman Functions -----

def calculate_public_keys():
    try:
        p = int(p_entry.get())
        g = int(g_entry.get())
        aarav_private = int(aarav_private_entry.get())
        vihaan_private = int(vihaan_private_entry.get())

        # Calculate public keys
        aarav_public = (g ** aarav_private) % p
        vihaan_public = (g ** vihaan_private) % p
```

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```
aarav_public_entry.delete(0, tk.END)
aarav_public_entry.insert(0, aarav_public)

vihaan_public_entry.delete(0, tk.END)
vihaan_public_entry.insert(0, vihaan_public)

status_label.config(
    text="Public keys generated successfully.",
    fg="green"
)

except ValueError:
    messagebox.showerror(
        "Invalid Input",
        "Please enter valid integer values."
    )

def calculate_shared_keys():
    try:
        p = int(p_entry.get())
        aarav_private = int(aarav_private_entry.get())
        vihaan_private = int(vihaan_private_entry.get())

        aarav_public = int(aarav_public_entry.get())
        vihaan_public = int(vihaan_public_entry.get())

        # Calculate shared secret keys
        aarav_shared = (vihaan_public ** aarav_private) % p
        vihaan_shared = (aarav_public ** vihaan_private) % p

        aarav_shared_entry.delete(0, tk.END)
        aarav_shared_entry.insert(0, aarav_shared)

        vihaan_shared_entry.delete(0, tk.END)
        vihaan_shared_entry.insert(0, vihaan_shared)
```

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```
status_label.config(
    text="Shared secret keys generated.",
    fg="blue"
)

except ValueError:
    messagebox.showerror(
        "Error",
        "Please generate public keys first."
    )

def verify_keys():
    try:
        aarav_shared = int(aarav_shared_entry.get())
        vihaan_shared = int(vihaan_shared_entry.get())

        if aarav_shared == vihaan_shared:
            result_label.config(
                text="✓ KEY EXCHANGE SUCCESSFUL\n"
                "Both users have the same secret key.",
                fg="green"
            )
        else:
            result_label.config(
                text="X KEY EXCHANGE FAILED\n"
                "The shared keys are different.",
                fg="red"
            )
    except ValueError:
        messagebox.showerror(
            "Error",
            "Please generate the shared keys first."
        )
```

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```
def clear_all():
    entries = [
        p_entry,
        g_entry,
        aarav_private_entry,
        vihaan_private_entry,
        aarav_public_entry,
        vihaan_public_entry,
        aarav_shared_entry,
        vihaan_shared_entry
    ]

    for entry in entries:
        entry.delete(0, tk.END)

    status_label.config(text="", fg="black")
    result_label.config(text="", fg="black")

# ----- Main Window -----

root = tk.Tk()
root.title("Diffie-Hellman Key Exchange")
root.geometry("750x650")
root.resizable(False, False)
root.configure(bg="#f2f4f7")

# ----- Title -----

title = tk.Label(
    root,
    text="Diffie-Hellman Key Exchange",
    font=("Arial", 22, "bold"),
```

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```
        bg="#f2f4f7",
        fg="#1f2937"
    )
    title.pack(pady=15)

    subtitle = tk.Label(
        root,
        text="Secure Shared Key Generation",
        font=("Arial", 11),
        bg="#f2f4f7",
        fg="#6b7280"
    )
    subtitle.pack()

# ----- Public Parameters -----

    public_frame = tk.LabelFrame(
        root,
        text=" Public Parameters ",
        font=("Arial", 12, "bold"),
        bg="white",
        padx=20,
        pady=15
    )
    public_frame.pack(padx=30, pady=15, fill="x")

    tk.Label(
        public_frame,
        text="Prime Number (p):",
        font=("Arial", 11),
        bg="white"
    ).grid(row=0, column=0, padx=10, pady=8, sticky="w")

    p_entry = tk.Entry(
        public_frame,
```

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```
width=25,
font=("Arial", 11)
)
p_entry.grid(row=0, column=1, padx=10, pady=8)

tk.Label(
    public_frame,
    text="Primitive Root (g):",
    font=("Arial", 11),
    bg="white"
).grid(row=1, column=0, padx=10, pady=8, sticky="w")

g_entry = tk.Entry(
    public_frame,
    width=25,
    font=("Arial", 11)
)
g_entry.grid(row=1, column=1, padx=10, pady=8)

# ----- Users Frame -----

users_frame = tk.Frame(root, bg="#f2f4f7")
users_frame.pack(padx=30, pady=5, fill="x")

# ----- Aarav -----

aarav_frame = tk.LabelFrame(
    users_frame,
    text=" Aarav ",
    font=("Arial", 12, "bold"),
    bg="white",
    padx=15,
    pady=15
)
```

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```
aarav_frame.pack(side="left", padx=10, fill="both", expand=True)
```

```
tk.Label(  
    aarav_frame,  
    text="Private Key:",  
    font=("Arial", 10),  
    bg="white"  
).pack(anchor="w", pady=5)
```

```
aarav_private_entry = tk.Entry(  
    aarav_frame,  
    font=("Arial", 11)  
)  
aarav_private_entry.pack(fill="x")
```

```
tk.Label(  
    aarav_frame,  
    text="Public Key:",  
    font=("Arial", 10),  
    bg="white"  
).pack(anchor="w", pady=(15, 5))
```

```
aarav_public_entry = tk.Entry(  
    aarav_frame,  
    font=("Arial", 11)  
)  
aarav_public_entry.pack(fill="x")
```

```
tk.Label(  
    aarav_frame,  
    text="Shared Secret:",  
    font=("Arial", 10),  
    bg="white"  
).pack(anchor="w", pady=(15, 5))
```

```
aarav_shared_entry = tk.Entry(  
    aarav_frame,  
    font=("Arial", 11)  
)  
aarav_shared_entry.pack(fill="x")
```

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```
    aarav_frame,
    font=("Arial", 11)
)
aarav_shared_entry.pack(fill="x")

# ----- Vihaan -----

vihaan_frame = tk.LabelFrame(
    users_frame,
    text=" Vihaan ",
    font=("Arial", 12, "bold"),
    bg="white",
    padx=15,
    pady=15
)
vihaan_frame.pack(side="right", padx=10, fill="both", expand=True)

tk.Label(
    vihaan_frame,
    text="Private Key:",
    font=("Arial", 10),
    bg="white"
).pack(anchor="w", pady=5)

vihaan_private_entry = tk.Entry(
    vihaan_frame,
    font=("Arial", 11)
)
vihaan_private_entry.pack(fill="x")

tk.Label(
    vihaan_frame,
    text="Public Key:",
    font=("Arial", 10),
    bg="white"
```

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```
).pack(anchor="w", pady=(15, 5))

vihaan_public_entry = tk.Entry(
    vihaan_frame,
    font=("Arial", 11)
)
vihaan_public_entry.pack(fill="x")

tk.Label(
    vihaan_frame,
    text="Shared Secret:",
    font=("Arial", 10),
    bg="white"
).pack(anchor="w", pady=(15, 5))

vihaan_shared_entry = tk.Entry(
    vihaan_frame,
    font=("Arial", 11)
)
vihaan_shared_entry.pack(fill="x")

# ----- Buttons -----

button_frame = tk.Frame(root, bg="#f2f4f7")
button_frame.pack(pady=18)

tk.Button(
    button_frame,
    text="Calculate Public Keys",
    command=calculate_public_keys,
    font=("Arial", 10, "bold"),
    width=20,
    padx=5,
    pady=8
).grid(row=0, column=0, padx=5)
```

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```
tk.Button(  
    button_frame,  
    text="Generate Shared Keys",  
    command=calculate_shared_keys,  
    font=("Arial", 10, "bold"),  
    width=20,  
    padx=5,  
    pady=8  
)grid(row=0, column=1, padx=5)
```

```
tk.Button(  
    button_frame,  
    text="Verify Key Exchange",  
    command=verify_keys,  
    font=("Arial", 10, "bold"),  
    width=20,  
    padx=5,  
    pady=8  
)grid(row=0, column=2, padx=5)
```

```
tk.Button(  
    button_frame,  
    text="Clear",  
    command=clear_all,  
    font=("Arial", 10, "bold"),  
    width=15,  
    padx=5,  
    pady=8  
)grid(row=1, column=1, pady=10)
```

```
# ----- Status -----
```

```
status_label = tk.Label(  
    root,
```

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```
text="",
font=("Arial", 10),
bg="#f2f4f7"
)
status_label.pack()

result_label = tk.Label(
    root,
    text="",
    font=("Arial", 13, "bold"),
    bg="#f2f4f7",
    justify="center"
)
result_label.pack(pady=10)

# ----- Start Application -----

root.mainloop()
```

Output

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